

VENN DIAGRAM_25-9-2026

Question Type : MCQ		Marks For Correct Answer: 3 Negative Mark: 0
Directions for questions 1 to 3: These questions are based on the following information: To cash in on the summer season, a businessman started a shop in an office building to sell water packets and buttermilk packets. On the first day, he observed the following: i. The number of persons who bought neither water nor buttermilk and the number of persons who bought water are in the ratio 1 : 3. ii. The number of persons who bought both and the number of persons who bought neither are in the ratio 5 : 2. iii. Only half of the persons who bought buttermilk also bought water. iv. 25 persons bought only water.		Question No. 1 What is the total number of persons visiting the shop on the first day? ☐ 450 ☐ 275 ☐ 325 ☐ 300
Question Type : MCQ		Marks For Correct Answer: 3 Negative Mark: 0
Directions for questions 1 to 3: These questions are based on the following information: To cash in on the summer season, a businessman started a shop in an office building to sell water packets and buttermilk packets. On the first day, he observed the following: i. The number of persons who bought neither water nor buttermilk and the number of persons who bought water are in the ratio 1 : 3. ii. The number of persons who bought both and the number of persons who bought neither are in the ratio 5 : 2. iii. Only half of the persons who bought buttermilk also bought water. iv. 25 persons bought only water.		Question No. 2 What is the number of persons who bought either water or buttermilk? ☐ 450 ☐ 275 ☐ 400 ☐ 300
Question Type : MCQ		Marks For Correct Answer: 3 Negative Mark: 0
Directions for questions 1 to 3: These questions are based on the following information: To cash in on the summer season, a businessman started a shop in an office building to sell water packets and buttermilk packets. On the first day, he observed the following: i. The number of persons who bought neither water nor buttermilk and the number of persons who bought water are in the ratio 1 : 3. ii. The number of persons who bought both and the number of persons who bought neither are in the ratio 5 : 2. iii. Only half of the persons who bought buttermilk also bought water. iv. 25 persons bought only water.		Question No. 3 What is the number of persons who did not buy buttermilk? ☐ 50 ☐ 75 ☐ 200 ☐ 25
Question Type : MCQ		Marks For Correct Answer: 3 Negative Mark: 0
Directions for questions 4 to 7: These questions are based on the following information. The following details are known about the travel choices of 600 residents of an apartment complex. i. 170 people travel only by a two wheeler. ii. 20 people daily commute only by bus, which is one-fourth of the number of people who commute by all the three modes i.e. bus, car and a two wheeler. iii. The number of people who travel only by car is 100 less than those who travel by only two wheeler. iv. The number of people who travel by bus and car is the same as that of bus and two wheeler. v. The number of people who travel by bus and exactly one other mode is equal to the number of people who travel by only car and two wheeler. vi. 60 residents do not travel by any of the three modes of transport mentioned.		Question No. 4 What is the percentage of residents who travel by exactly two modes of transport? ☐ 30% ☐ 37% ☐ $33\frac{1}{3}\%$ ☐ $33\frac{2}{3}\%$

Directions for questions 4 to 7: These questions are based on the following information.

The following details are known about the travel choices of 600 residents of an apartment complex.

- 170 people travel only by a two wheeler.
- 20 people daily commute only by bus, which is one-fourth of the number of people who commute by all the three modes i.e. bus, car and a two wheeler.
- The number of people who travel only by car is 100 less than those who travel by only two wheeler.
- The number of people who travel by bus and car is the same as that of bus and two wheeler.
- The number of people who travel by bus and exactly one other mode is equal to the number of people who travel by only car and two wheeler.
- 60 residents do not travel by any of the three modes of transport mentioned.

Question No. 5

What is the number of residents who travel by car?

☐ 200
 ☐ 300
 ☐ 400
 ☐ 350

Directions for questions 4 to 7: These questions are based on the following information.

The following details are known about the travel choices of 600 residents of an apartment complex.

- 170 people travel only by a two wheeler.
- 20 people daily commute only by bus, which is one-fourth of the number of people who commute by all the three modes i.e. bus, car and a two wheeler.
- The number of people who travel only by car is 100 less than those who travel by only two wheeler.
- The number of people who travel by bus and car is the same as that of bus and two wheeler.
- The number of people who travel by bus and exactly one other mode is equal to the number of people who travel by only car and two wheeler.
- 60 residents do not travel by any of the three modes of transport mentioned.

Question No. 6

What is the number of residents who travel by car or bus but not a two wheeler?

☐ 370
 ☐ 140
 ☐ 170
 ☐ 200

Directions for questions 4 to 7: These questions are based on the following information.

The following details are known about the travel choices of 600 residents of an apartment complex.

- 170 people travel only by a two wheeler.
- 20 people daily commute only by bus, which is one-fourth of the number of people who commute by all the three modes i.e. bus, car and a two wheeler.
- The number of people who travel only by car is 100 less than those who travel by only two wheeler.
- The number of people who travel by bus and car is the same as that of bus and two wheeler.
- The number of people who travel by bus and exactly one other mode is equal to the number of people who travel by only car and two wheeler.
- 60 residents do not travel by any of the three modes of transport mentioned.

Question No. 7

What is the number of residents who do not travel by car?

☐ 300
 ☐ 140
 ☐ 240
 ☐ 70

Directions for questions 8 to 11: These questions are based on the following information.

An exam is conducted every week which is of three sections: maths, physics and chemistry. On a particular day, when all the students attended the exam, 30% students cleared cutoffs in only maths and chemistry which is thrice that of students who cleared cutoffs in only maths and physics, which is half of that of students who cleared cutoffs in only physics and chemistry.

The numbers of students who cleared cutoff in only maths is the same as that of only physics which is same as that of only chemistry, which is half of those who cleared cutoffs in all the three sections.

Every student has cleared the cutoff in at least one section.

Question No. 8

What is the ratio of number of students who cleared cutoffs in maths to that of students who cleared cutoff in physics?

☐ 32 : 27
 ☐ 32 : 37
 ☐ 27 : 32
 ☐ Cannot be determined

Question Type : MCQ Directions for questions 8 to 11: These questions are based on the following information. An exam is conducted every week which is of three sections: maths, physics and chemistry. On a particular day, when all the students attended the exam, 30% students cleared cutoffs in only maths and chemistry which is thrice that of students who cleared cutoffs in only maths and physics, which is half of that of students who cleared cutoffs in only physics and chemistry. The numbers of students who cleared cutoff in only maths is the same as that of only physics which is same as that of only chemistry, which is half of those who cleared cutoffs in all the three sections. Every student has cleared the cutoff in at least one section.	Marks For Correct Answer: 3 Negative Mark: 1 Question No. 9 What is the percentage of students who cleared cutoffs in at least two sections? ☐ 70% ☐ 75% ☐ 76% ☐ 80%
Question Type : MCQ Directions for questions 8 to 11: These questions are based on the following information. An exam is conducted every week which is of three sections: maths, physics and chemistry. On a particular day, when all the students attended the exam, 30% students cleared cutoffs in only maths and chemistry which is thrice that of students who cleared cutoffs in only maths and physics, which is half of that of students who cleared cutoffs in only physics and chemistry. The numbers of students who cleared cutoff in only maths is the same as that of only physics which is same as that of only chemistry, which is half of those who cleared cutoffs in all the three sections. Every student has cleared the cutoff in at least one section.	Marks For Correct Answer: 3 Negative Mark: 1 Question No. 10 What is the minimum possible number for the total number of students who have appeared for the exam? ☐ 25 ☐ 50 ☐ 100 ☐ 625
Question Type : MCQ Directions for questions 8 to 11: These questions are based on the following information. An exam is conducted every week which is of three sections: maths, physics and chemistry. On a particular day, when all the students attended the exam, 30% students cleared cutoffs in only maths and chemistry which is thrice that of students who cleared cutoffs in only maths and physics, which is half of that of students who cleared cutoffs in only physics and chemistry. The numbers of students who cleared cutoff in only maths is the same as that of only physics which is same as that of only chemistry, which is half of those who cleared cutoffs in all the three sections. Every student has cleared the cutoff in at least one section.	Marks For Correct Answer: 3 Negative Mark: 1 Question No. 11 Which section of the exam has the highest number of students that cleared the cutoff? ☐ Maths ☐ Physics ☐ Chemistry ☐ All the three sections
Question Type : MCQ Directions for questions 12 to 15: These questions are based on the following information. A school offers courses in music and painting as extracurricular activities to its students. 250 students participate in music, 350 students participate in painting and 150 participate in both. When the school had introduced two more courses i.e. dance and movie-making, it is observed that among each group of students viz., groups of students who participate in music alone, painting alone and both music and painting, the numbers of students who opted for only dance additionally, for only movie-making additionally, for both dance and movie-making additionally and for no additional course are in the ratio 1: 2: 3: 4. Later, from among each group of students who were enrolled in music, painting and at least one additional course, one-third of them have opted out of both music and painting.	Marks For Correct Answer: 3 Negative Mark: 1 Question No. 12 What is the number of students who participate in exactly two out of the four courses? ☐ 165 ☐ 150 ☐ 215 ☐ 320

Question Type : MCQ Directions for questions 12 to 15: These questions are based on the following information. A school offers courses in music and painting as extracurricular activities to its students. 250 students participate in music, 350 students participate in painting and 150 participate in both. When the school had introduced two more courses i.e. dance and movie-making, it is observed that among each group of students viz., groups of students who participate in music alone, painting alone and both music and painting, the numbers of students who opted for only dance additionally, for only movie-making additionally, for both dance and movie-making additionally and for no additional course are in the ratio 1 : 2 : 3 : 4. Later, from among each group of students who were enrolled in music, painting and at least one additional course, one-third of them have opted out of both music and painting.	Marks For Correct Answer: 3 Negative Mark: 1 Question No. 13 What is the number of students who participate in all the four courses? ☐ 60 ☐ 45 ☐ 150 ☐ 30
Question Type : MCQ Directions for questions 12 to 15: These questions are based on the following information. A school offers courses in music and painting as extracurricular activities to its students. 250 students participate in music, 350 students participate in painting and 150 participate in both. When the school had introduced two more courses i.e. dance and movie-making, it is observed that among each group of students viz., groups of students who participate in music alone, painting alone and both music and painting, the numbers of students who opted for only dance additionally, for only movie-making additionally, for both dance and movie-making additionally and for no additional course are in the ratio 1 : 2 : 3 : 4. Later, from among each group of students who were enrolled in music, painting and at least one additional course, one-third of them have opted out of both music and painting.	Marks For Correct Answer: 3 Negative Mark: 1 Question No. 14 What is the total number of students who have opted out of any course? ☐ 50 ☐ 45 ☐ 30 ☐ 40
Question Type : MCQ Directions for questions 12 to 15: These questions are based on the following information. A school offers courses in music and painting as extracurricular activities to its students. 250 students participate in music, 350 students participate in painting and 150 participate in both. When the school had introduced two more courses i.e. dance and movie-making, it is observed that among each group of students viz., groups of students who participate in music alone, painting alone and both music and painting, the numbers of students who opted for only dance additionally, for only movie-making additionally, for both dance and movie-making additionally and for no additional course are in the ratio 1 : 2 : 3 : 4. Later, from among each group of students who were enrolled in music, painting and at least one additional course, one-third of them have opted out of both music and painting.	Marks For Correct Answer: 3 Negative Mark: 1 Question No. 15 What is the number of students who participate in movie-making course? ☐ 220 ☐ 320 ☐ 190 ☐ 225